Ethical and legal declarations

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Research ethics

The present secondary analysis was exempt from additional ethical review under Article 32, item (1), of China's Measures for Ethical Review of Life Science and Medical Research Involving Human Participants (2023). It used lawfully obtained publicly available data, caused no harm to participants, and did not involve sensitive personal information or commercial interests. The original Kumar data collection reported University of Texas at Austin approval under protocol 2020-03-0073. The reviewed BNCI2014-001 source publication and dataset description did not supply a dataset-2a-specific committee, approval number or consent statement; their data-collection ethics details are not inferred from another dataset. Ziyuan Zhu had no access to identifying participant information and made no attempt to re-identify participants.

Informed consent

The original Kumar study reported written informed consent before data collection. No dataset-2a-specific consent statement was identified in the reviewed BNCI source documents. The present study analyzed coded public records; no new participants were recruited or contacted and no additional consent was collected.

Author contributions

The author list and contributions below retain the previously confirmed information. Final author approval of this expanded manuscript has not yet been recorded; the earlier approval applies to its earlier version.

Ziyuan Zhu: Conceptualization; Methodology; Software; Formal analysis; Validation; Visualization; Writing – original draft; Writing – review & editing.

Sihao Liu assisted with code verification, organization of project files, and checking the journal's submission policies.

Haoyang Yu assisted with data checking and aggregation, project-file organization, and GitHub uploads.

Use of large language models, AI and machine learning tools

OpenAI Codex assisted with code revision, data analysis and result checks, manuscript preparation, GitHub updates and file organization; OpenAI ChatGPT assisted with grammatical correction. The authors are responsible for verifying sources, scientific claims and the final manuscript.

Conflict of interest

The authors state no conflict of interest.

Research funding

None declared.

Data availability

Kumar2024 source EEG is available through NEMAR nm000177/v1.0.5, https://doi.org/10.82901/nemar.nm000177.v1.0.5. BNCI2014-001 original MAT recordings are available through https://bnci-horizon-2020.eu/database/data-sets/001-2014/ under CC BY-ND 4.0; NEMAR nm000139/v1.0.3 provides their versioned original byte hashes. The project repository https://github.com/jackzhu119/few-shot-mi-eeg, branch `research/paper2-alignment-integrated-revision`, supplies the isolated revision's input manifests, saved-prediction lineage, derived participant/session tables, figure data, analysis code and reproduction instructions. Historical predictions and snapshots remain at their fixed original paths. Cross-project source and overlap ledgers distinguish tools and repeated analyses from independent datasets. Original EEG and third-party materials retain their source licenses.

Software availability

Original analysis code is available at https://github.com/jackzhu119/few-shot-mi-eeg under the MIT license. Original derived data, figures and manuscript content use CC BY 4.0. The adapted citation style retains its CC BY-SA 3.0 license.